SECTION A: MULTIPLE CHOICE QUESTIONS - 13 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. START WITH SECTION A AND HAND THE OPTIC READER FORM IN AFTER 75 MINUTES.
2. First circle your answers on this paper and only mark the answers on the form when you are certain of your choice. You may not erase on the optic reader form.
3. Do all scribbling on the facing page. It will not be marked.
4. Use a soft pencil to write on the optic reader form.
5. Use **SIDE 2** of the optic reader form.
   Fill in your personal information in pencil on side 2.
   Fill in your student number from top to bottom and then code it.
6. **You may start with section B as soon as you have marked your answers on the optic reader form.**

AFDELING A : MEERVOUDIGE KEUSEVRAE - 13 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. BEGIN MET AFDELING A EN HANDIG DIE MERKLEESVORM IN NA 75 MINUTE.
2. Omkring jou antwoorde eers in hierdie vraestel en merk die antwoorde op die merkleesvorm eers as jy seker is van jou keuse. Jy mag nie uitvee op die merkleesvorm nie.
3. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
4. Gebruik 'n sagte potlood om op die merkleesvorm te skryf.
5. Gebruik **KANT 2** van die merkleesvorm.
   Vul u persoonlike inligting in potlood in op kant 2.
   Vul u studentenommer van bo na onder in en kodeer dit.
6. **U kan met afdeling B begin sodra u die antwoorde op die merkleesvorm ingevul het.**

Question 1 / Vraag 1

If / Indien $|x^2+1| \geq 2$ then / dan is

(a)	$x \in [-1,1]$	(b)	$x \in [-2,2]$	(c)	$x \geq 1$ and/en $x \leq -1$
(d)	$x \in (-\infty,-1] \cup [1,\infty)$	(e)	None of these / Geen van hierdie		

Question 2 / Vraag 2

If / Indien $|\ln x - 1| = 1$ then / dan is

(a)	$x = 1$ or /of $x = e - 1$	(b)	$x = 1$ or /of $x = e^2$	(c)	$x = 1$ or /of $x = e$
(d)	$x = 0$ or /of $x = e - 1$	(e)	$x = 0$	(f)	None of these / Geen van hierdie

Question 3 / Vraag 3

If $|x-3| = |2x+1|$, then

Indien $|x-3| = |2x+1|$, dan is

(a)	$x = \frac{1}{3}$ or/of $x = \frac{2}{3}$	(b)	$x = \frac{2}{3}$ or/of $x = -\frac{1}{4}$	(c)	$x = \frac{2}{3}$ or/of $x = -4$
(d)	$x = \frac{2}{3}$ or/of $x = -\frac{2}{3}$	(e)	$x = \frac{1}{3}$ or/of $x = \frac{3}{4}$	(f)	None of these / Geen van hierdie

Question 4 / Vraag 4

Let / Laat

$$f(x) = \begin{cases} 2 & \text{if / as } \ |x| \leq 1 \\ 1-x & \text{if / as } \ |x| > 1 \end{cases}$$

For the piece-wise defined function which statement is true?

Vir die stukswysgedefinieerde funksie watter bewering is waar?

(a)	The domain of f is $[0,\infty)$ and the range is $\mathbb{R}$
	Die definisieversameling van f is $[0,\infty)$ en die waardeversameling van f is $\mathbb{R}$
(b)	The domain of f is $\mathbb{R}$ and the range is $(-\infty,0) \cup [2,\infty)$
	Die definisieversameling van f is $\mathbb{R}$ en die waardeversameling van f is $(-\infty,0) \cup [2,\infty)$
(c)	The domain of f is $\mathbb{R}$ and the range is $\mathbb{R}\backslash(0,2]$
	Die definisieversameling van f is $\mathbb{R}$ en die waardeversameling van f is $\mathbb{R}\backslash(0,2]$
(d)	The domain of f is $[0,\infty)$ and the range is $(-\infty,2]$
	Die definisieversameling van f is $[0,\infty)$ en die waardeversameling van f is $(-\infty,2]$
(e)	None of these / Geen van hierdie

Question 5 / Vraag 5

$3\ln(e^{2x}(e^x)^3) + 2e^{\ln 1} =$

(a)	$15x+2$	(b)	$15x$	(c)	4
(d)	1	(e)	0	(f)	None of these / Geen van hierdie

Question 6 / Vraag 6

$\tan(-\frac{17\pi}{6}) =$

(a)	$-\sqrt{3}$	(b)	-1	(c)	$\frac{1}{\sqrt{3}}$
(d)	$\sqrt{3}$	(e)	$-\frac{1}{\sqrt{3}}$	(f)	None of these / Geen van hierdie

Question 7 / Vraag 7

$\sec(\frac{14\pi}{2}) =$

(a)	1	(b)	-1	(c)	0
(d)	$\frac{\sqrt{3}}{2}$	(e)	$-\frac{2}{\sqrt{2}}$	(f)	None of these / Geen van hierdie

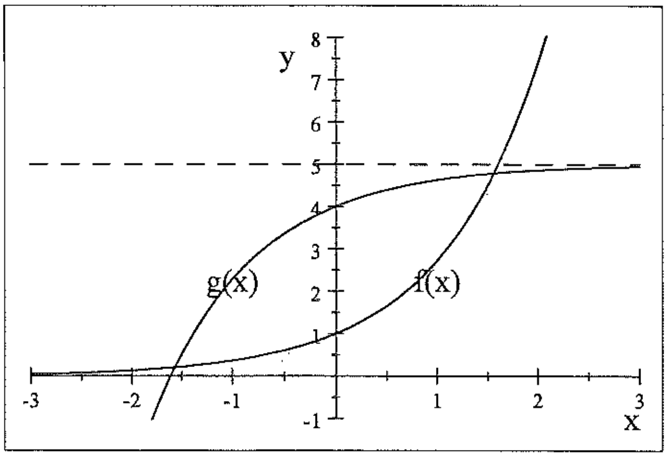
Question 8 / Vraag 8

A function f has a domain $[-1,1]$ and range $[0,2]$. The inverse g^{-1} of the function $g(x) = f(x-1)+1$ has

'n Funksie f het as definisieversameling $[-1,1]$ en waardeversameling $[0,2]$. Die inverse g^{-1} van die funksie $g(x) = f(x-1)+1$ het

(a)	Domain / Definisieversameling: $[0,2]$, Range / Waardeversameling: $[1,3]$
(b)	Domain / Definisieversameling: $[1,3]$, Range / Waardeversameling: $[0,2]$
(c)	Domain / Definisieversameling: $[1,3]$, Range / Waardeversameling: $[-2,0]$
(d)	Domain / Definisieversameling: $[-2,0]$, Range / Waardeversameling: $[0,3]$
(e)	None of these / Geen van hierdie

Question 9 / Vraag 9



The function $f(x) = e^x$ and the transformed function $g(x)$ are shown in the sketch. The function $g(x)$ is given by

Die funksie $f(x) = e^x$ en die getransformeerde funksie $g(x)$ word beide in die skets getoon. Die funksie $g(x)$ word gegee deur

(a)	$g(x) = 5 - e^x$
(b)	$g(x) = e^{-x} + 5$
(c)	$g(x) = -e^x + 5$
(d)	$g(x) = 5 - e^{-x}$
(e)	None of these / Geen van hierdie

Question 10 / Vraag 10

The domain of the function $f(x) = \frac{4}{x}\ln(4-x^2)$ is

Die definisieversameling van die funksie $f(x) = \frac{4}{x}\ln(4-x^2)$ is

(a)	$(-\infty,0) \cup (0,\infty)$	(b)	$x \leq \pm 2, x \neq 0$	(c)	$(-2,0)\cup(0,2)$
(d)	None of these / Geen van hierdie				

Question 11 / Vraag 11

The solution of $\frac{x^2-4x+3}{x} > 0$ is

Die oplossing van $\frac{x^2-4x+3}{x} > 0$ is

(a)	$x \in (0,1)\cup(3,\infty)$	(b)	$x \in (-\infty,1)\cup(3,\infty)$	(c)	$x \in (-\infty,0)\cup(1,3)$
(d)	$x \in (-4,-2)$	(e)	None of these / Geen van hierdie		

Question 12 / Vraag 12

The solution of $2\sin x + 1 = 0$ on $[-2\pi, 2\pi]$ is

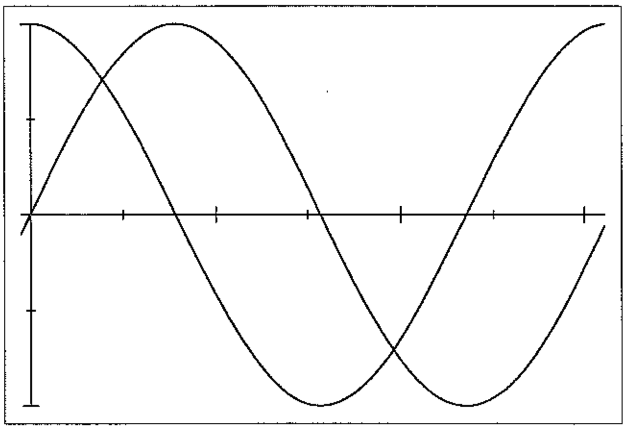
Die oplossing van $2\sin x + 1 = 0$ op $[-2\pi, 2\pi]$ is

(a)	$x \in \{\frac{5\pi}{6}, \frac{7\pi}{6}, -\frac{5\pi}{6}, -\frac{7\pi}{6}\}$	(b)	$x = \pm\frac{7\pi}{6}$ or $x = \pm\frac{11\pi}{6}$	(c)	$x \in \{\frac{7\pi}{6}, \frac{11\pi}{6}, -\frac{\pi}{6}, -\frac{5\pi}{6}\}$
(d)	$x = \frac{7\pi}{6} + 2k\pi$ or / of $x = \frac{11\pi}{6} + 2k\pi$	(e)	None of these / Geen van hierdie		

Question 13 / Vraag 13

The graphs of $y = \sin x$ and $y = \cos x$ on $[0, 2\pi]$ are shown on the same set of axes.

Die grafieke van $y = \sin x$ en $y = \cos x$ op $[0, 2\pi]$ word op dieselfde assestelsel getoon.



The solution of $\sin x > \cos x$ is

Die oplossing van $\sin x > \cos x$ is

(a)	$x \in (\frac{\pi}{4}, \frac{5\pi}{4})$	(b)	$x \in (0,\frac{\pi}{4})\cup(\frac{5\pi}{4},\frac{7\pi}{4})$	(c)	$x \in (\frac{\pi}{4}, \frac{7\pi}{4})$
(d)	$x \in (\frac{\pi}{2}, \pi)$	(e)	None of these / Geen van hierdie		

SECTION B: 27 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. Answer the following questions on this paper **in ink.**
2. Work done in pencil or in red ink will not be marked.
3. Do not use correction fluid ("Tipp-Ex").
4. Do all scribbling on the facing page. It will not be marked.
   If you need more space for an answer, use the facing page and indicate it clearly by framing it.
5. Show all steps and calculations and explain your answers.

AFDELING B: 27 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. Beantwoord die volgende vrae op hierdie vraestel **in ink.**
2. Werk wat in potlood of rooi ink beantwoord is, word nie nagesien nie.
3. Korrigeer-vloeistof ("Tipp-Ex") mag nie gebruik word nie.
4. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
   As jy nog ruimte vir 'n antwoord nodig het, gebruik die teenblad en dui dit duidelik aan **deur 'n raam daarom te trek.**
5. Toon alle stappe en berekeninge en verduidelik jou antwoorde.

Question 1 / Vraag 1

Fill in the **answer only** in the space provided. Rough work can be done on the opposing page and will not be marked.

Vul **slegs die antwoord** in die spasie in. Rofwerk kan op die teenblad gedoen word en sal nie gemerk word nie.

1.1 Given that $f(x) = x^2 + 1$, $g(x) = \frac{1}{x-2}$ and $h(x) = \sqrt{x}$, find the following:

Gegee dat $f(x) = x^2 + 1$, $g(x) = \frac{1}{x-2}$ en $h(x) = \sqrt{x}$, vind die volgende:

(a) $\left(\frac{h}{g}\right)(x) =$ $\sqrt{x}(x-2)$ 1

(b) $(g \circ f)(x) =$ $\frac{1}{x^2-1}$ 1

1.2 Given that $H = f \circ g \circ h$. If $H(x) = \sqrt[4]{\sqrt{x} - 4}$ and $h(x) = \sqrt{x}$, then

Gegee dat $H = f \circ g \circ h$. Indien $H(x) = \sqrt[4]{\sqrt{x} - 4}$ en $h(x) = \sqrt{x}$, dan is

(a) $f(x) =$ $\sqrt[4]{x^1} = x^{1/4}$

(b) $g(x) =$ $x - 4$

1

1.3 If / Indien $f(x) = e^x$ and / en $g(x) = \ln x$ then / dan is

$f(g(x)) = x$ for / vir $x \in$ $(0, \infty)$

$g(f(x)) = x$ for / vir $x \in$ $\mathbb{R}$

1

Question 2 / Vraag 2

Given that / Gegee dat $f(x) = \frac{x-1}{x+1}$, determine / bepaal $f^{-1}(x)$.

$y = \frac{x-1}{x+1}$ Consider $x = \frac{y-1}{y+1}$

$\Rightarrow xy + x = y - 1$

$\Rightarrow y(x-1) = -(x+1)$

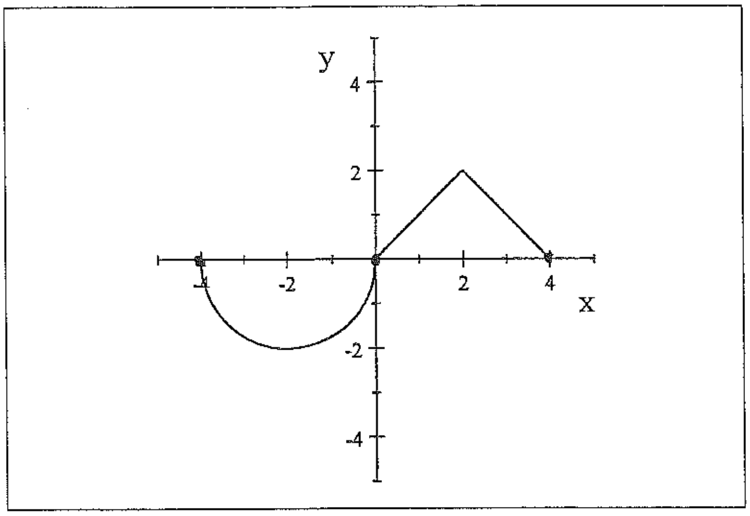
$\Rightarrow y = \frac{-(x+1)}{x-1}$

2

So $f^{-1}(x) = \frac{-(x+1)}{x-1}$

Question 3 / Vraag 3

Given the graph of the function $f(x)$ below. Gegee die grafiek van $f(x)$ hieronder.



Write down the equation of $f(x)$. Skryf die vergelyking van $f(x)$ neer.

$$f(x) = \begin{cases} -\sqrt{4-(x+2)^2} & \text{if } -4 \le x \le 0 \\ -|x-2|+2 & \text{if } 0 < x \le 4 \end{cases}$$

or

$$\begin{cases} -\sqrt{4-(x+2)^2} & \text{if } -4 \le x \le 0 \\ x & \text{if } 0 < x \le 2 \\ -x+4 & \text{if } 2 < x \le 4 \end{cases}$$

2

Question 4 / Vraag 4

Consider the functions $f(x) = x - 1$ and $g(x) = |x + 3|$. For which values of x will $f(g(x)) > 0$?

Beskou die funksies $f(x) = x - 1$ en $g(x) = |x + 3|$. Vir watter waardes van x is $f(g(x)) > 0$?

$$f(g(x)) = f(|x+3|)$$

$$= |x+3| - 1 > 0$$

$$\Rightarrow \quad |x+3| > 1$$

$$\Rightarrow \quad x+3 > 1 \quad \text{of} \quad x+3 < -1$$

$$x < -4$$

$$\Rightarrow \quad x > -2$$

3

Question 5 / Vraag 5

Solve the following: / Los die volgende op.

5.1 $\cos^2 x - \sin x \cos x = 0, \qquad x \in [0, 2\pi)$

$\cos x(\cos x - \sin x) = 0$

$\Rightarrow \cos x = 0 \quad \text{or} \quad \cos x = \sin x$

$\Rightarrow \tan x = 1$

$\Rightarrow x = \frac{\pi}{2} \text{ or } \frac{3\pi}{2} \quad \text{or} \quad x = \frac{\pi}{4} \text{ or } \frac{5\pi}{4}$

3

5.2 $2\ln x - \ln(x+1) = \ln 4 - \ln 3 \qquad ; \; x > 0$

$\ln x^2 - \ln(x+1) = \ln \frac{4}{3}$

$\Rightarrow \ln \frac{x^2}{x+1} = \ln \frac{4}{3}$

$\Rightarrow \frac{x^2}{x+1} = \frac{4}{3}$

$\Rightarrow 3x^2 - 4x - 4 = 0$

$\Rightarrow (3x+2)(x-2) = 0$

$\Rightarrow x = -\frac{2}{3} \quad \text{or} \quad x = 2$

n/a.

3

5.3 $\ln(3x-2)+\ln(x+1) \leq \ln 8$

$\ln(3x-2)(x+1) \leq \ln 8$ and $3x-2>0$

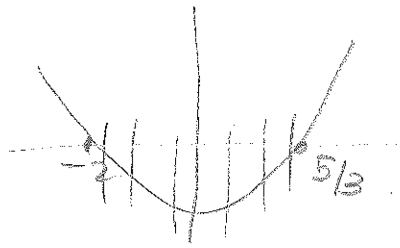
$\Rightarrow x > \frac{2}{3}$ (2)

$\Rightarrow (3x-2)(x+1) \leq 8$ and $x+1>0$

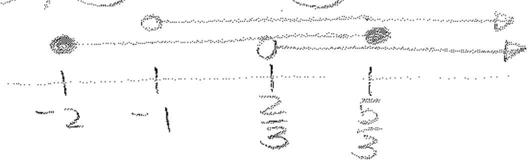
$\Rightarrow 3x^2+x-10 \leq 0$ $\Rightarrow x>-1$ (3)

$\Rightarrow (3x-5)(x+2) \leq 0$

$\Rightarrow x \in \left[-2, \frac{5}{3}\right]$ (1)



(1), (2) and (3)



So $x \in \left(\frac{2}{3}, \frac{5}{3}\right]$.

4

Question 6 / Vraag 6

6.1 Write down the definition of an even function. / Gee die definisie van 'n ewe funksie.

$f(-x) = f(x)$ for all x

1

6.2 Given $f(x) = x^3 - x$ and $g(x) = \sin x$. Is the function $h(x) = f(x).g(x)$ even, odd or neither? Motivate your answer.

Gegee $f(x) = x^3 - x$ en $g(x) = \sin x$. Is die funksie $h(x) = f(x).g(x)$ ewe, onewe of nie een van die twee nie? Motiveer jou antwoord.

$h(x) = (x^3-x)\sin x$

$h(-x) = ((-x)^3-(-x))\sin(-x)$

$= (-x^3+x)(-\sin(x))$

$= (x^3-x)(\sin x) = h(x)$.

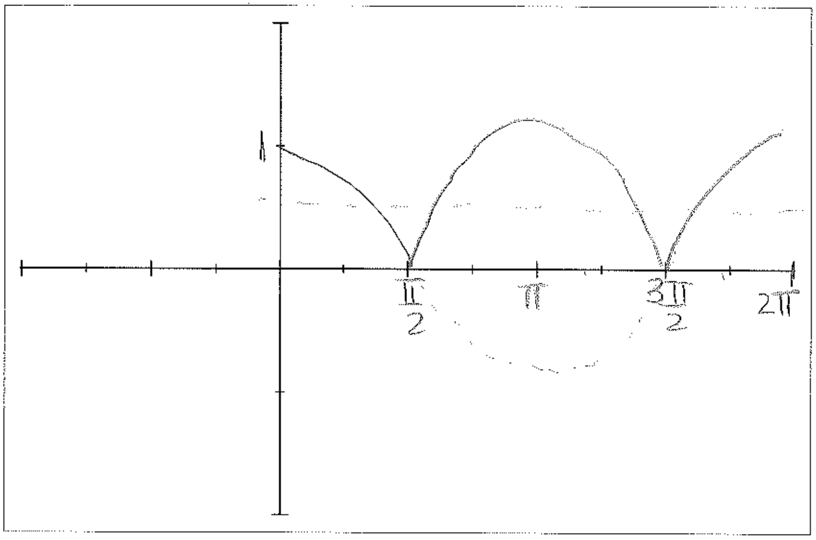
$\Rightarrow h(x)$ is an even function

2

Question 5 / Vraag 5

5.1 Sketch the graph (in ink) of $y = |\cos x|$ for $x \in [0, 2\pi]$.
(Remember to indicate all the relevent information on the graph.)

Skets die grafiek (in ink) van $y = |\cos x|$ vir $x \in [0, 2\pi]$.
(Onthou om alle belangrike inligting op die grafiek aan te dui.)



5.2 Use the graph to solve for x if $2|\cos x| > 1$.

Gebruik die grafiek en los vir x op indien $2|\cos x| > 1$.

$|\cos x| > \frac{1}{2}$

Consider $\cos x = \frac{1}{2}$. RA $x = \frac{\pi}{3}$ (1°)

$x = \frac{2\pi}{3}$ (2°)

$x = \frac{4\pi}{3}$ (3rd)

$x = \frac{5\pi}{3}$ (4th)

So $x \in \left[0, \frac{\pi}{3}\right) \cup \left(\frac{2\pi}{3}, \frac{4\pi}{3}\right) \cup \left(\frac{5\pi}{3}, 2\pi\right]$.

3